STEEL BEAM / 01

Beam analysis.

Understand how your beam responds before the work begins.

> ! **For preliminary evaluation only.** This analysis is not a construction approval or a complete member design. Connections, bearing, stability and scaffold-specific requirements require separate review.

02 / VISUALIZE

SCHEMATIC

Beam arrangement

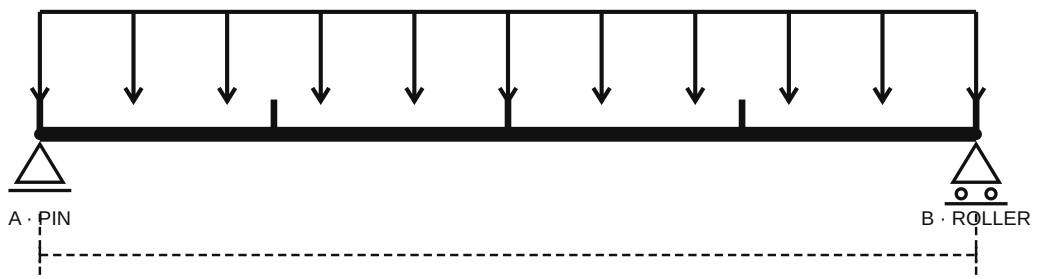
MODEL / 01

Beam geometry & loading

SCHEMATIC · NOT TO SCALE

ELEVATION · FULL-SPAN UNIFORM LOAD

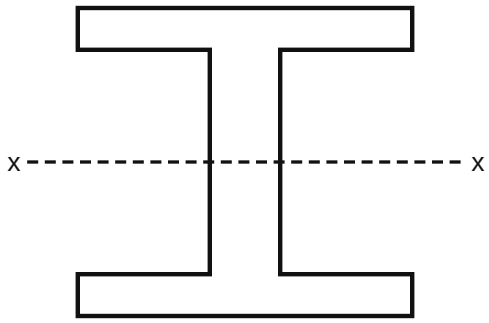
D 0.2 + L 0.8 kip/ft · self-weight included



L = 20 ft · DISTANCE BETWEEN SUPPORTS

BRACES / X 0, 5, 10, 15, 20 ft from A

SECTION · MAJOR-AXIS BENDING



STRONG AXIS · X–X

SECTION **W16X31**	07 / ELASTIC MODULUS E **29000 ksi**
08 / STRONG-AXIS I_x **375 in⁴**	09 / STIFFNESS FACTOR **0.8 × EI**

This illustration shows the supported one-span model only. Brace ticks describe lateral restraint; they do not add vertical supports. Section and stiffness values are properties, not dimensions. The optional load comparison is separate from the service-load diagram.

Analysis results

Responses for the load case shown above.

Service load case: 1.0D + 1.0L — results and diagrams below.

Support reaction · each	Maximum moment	Nominal EI deflection	Reduced EI deflection
10.31 kip	**51.55** kip-ft	**0.3413** in	**0.4266** in

Optional load comparison — illustrative only

Not a verified OSHA or ASD load combination. Not a capacity or compliance check. The selected comparison uses the user-proposed 2.5 / 1.67 = 1.49701 multiplier on the entered live load only. It includes entered dead load and selected beam self-weight once.

1.0D + (2.5 / 1.67)L = 0.2 + 0.031 + (2.5 / 1.67 × 0.8) = **1.4286 kip/ft**. For this uniform load on a simply supported beam: reaction at each support = **14.286 kip**; maximum moment = **71.43 kip-ft**.

The 1.67 divisor does not apply to every limit state, and the entered dead load may include applied loads that belong in the maximum intended load. The governing basis for the selected safety factor must be confirmed. Engineering review is required. The diagrams and deflections remain the separate 1.0D + 1.0L service case.

Calculation basis

Section W16X31 · Span 20 ft · E = 29,000 ksi · Ix = 375 in⁴. Nominal section properties and weight are from a third-party mirror of AISC Shapes Database v15; verify against the official source. The inputs are user-selected and the beam is simply supported with full-span uniform downward loading.

1.0D + 1.0L: 0.2 + 0.8 + 0.031 self-weight = **1.031 kip/ft**. Self-weight is included once.

R = wL/2 = 1.031 × 20 / 2 = 10.31 kip. M = wL²/8 = 51.55 kip-ft. Maximum bending deflection = 5wL⁴/(384EI), at 10 ft. The reduced value uses 0.8EI; the nominal value uses EI.

Lateral brace locations: 0, 5, 10, 15, 20 ft. Segment lengths: 5, 5, 5, 5 ft. Braces are not vertical supports.

Additional information required. Bending, shear, stability, bearing, connections and scaffold-specific capacity are not evaluated. No overall member adequacy is determined. Independent comparison not completed. The stiffness reduction is a selected standalone assumption, not a complete Direct Analysis Method. Shear deformation is excluded.

Project: Untitled project · Prepared by: Not specified · 2026-11-10T01:24:39.090Z · Engine 0.1.0

ⓘ **Independent comparison not completed**

Member capacity and scaffold-specific checks are not evaluated by this analysis.

Response diagrams

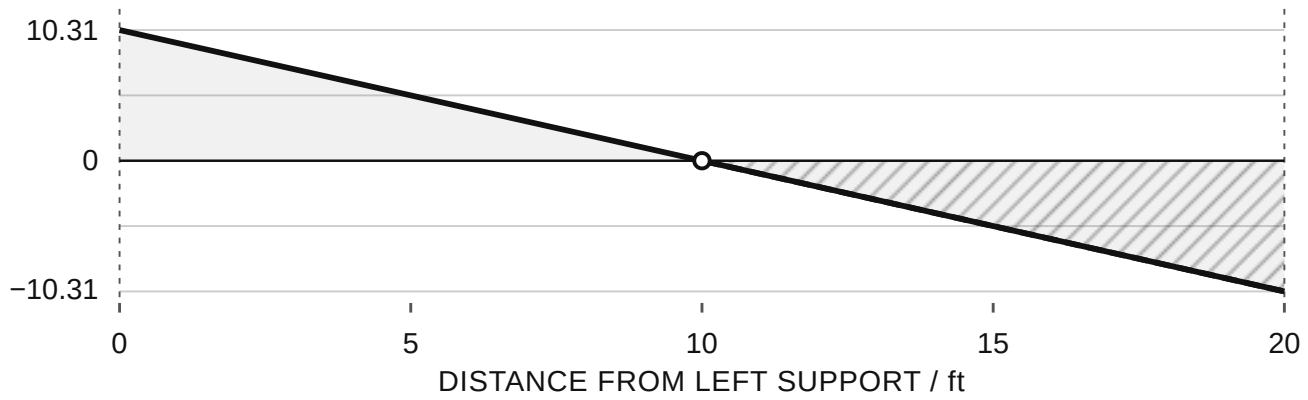
View how forces and displacement change along the span.

Service loads only. Positive shear equals the left reaction minus downward loads to the left of the cut; positive moment is sagging. Deflection is downward and uses reduced EI. Negative values use dashed lines and hatching. Curves connect sampled values from the response equations.

SERVICE CASE / 1.0D + 1.0L

Shear force

V / kip



MAXIMUM	MINIMUM
10.31 kip at 0 ft	**-10.31 kip at 20 ft**

Zero shear at 10 ft.

SERVICE CASE / 1.0D + 1.0L

Bending moment

M / kip-ft

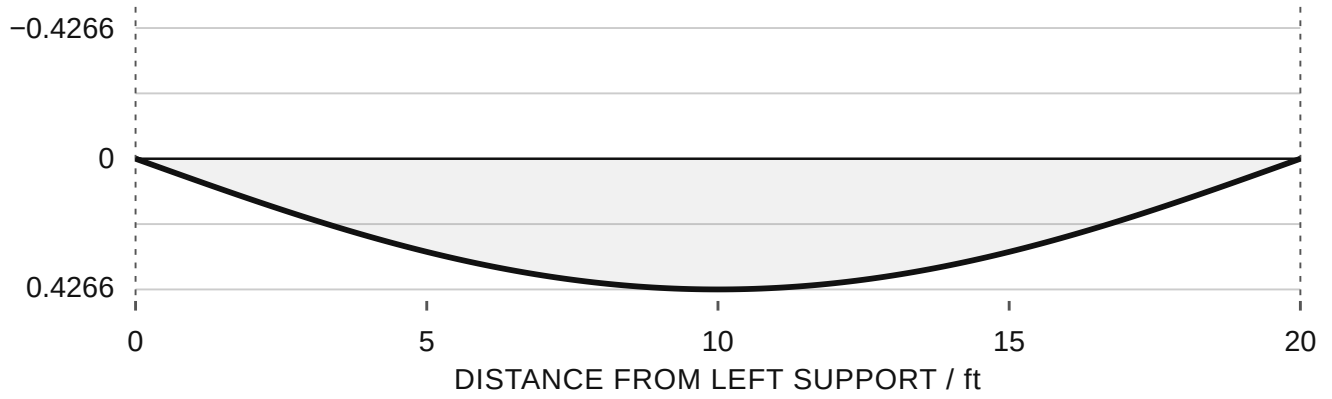


MAXIMUM	MINIMUM
51.55 kip-ft at 10 ft	**0 kip-ft at both supports**

SERVICE CASE / 1.0D + 1.0L

Bending deflection

δ / in



MAXIMUM	MINIMUM
0.4266 in at 10 ft	**0 in at both supports**

05 / REFERENCE

AISC v15 · mirrored

Section properties

AISC v15 properties from a third-party mirror; verify against the official source before use.

W **31** lb/ft	A **9.13** in²	d **15.9** in
bf **5.53** in	tw **0.275** in	tf **0.44** in
Ix **375** in⁴	Sx **47.2** in³	Zx **54** in³
Iy **12.4** in⁴	Sy **4.49** in³	Zy **7.03** in³
J **0.461** in⁴	Cw **739** in⁶	rx **6.41** in
ry **1.17** in	rts **1.42** in	ho **15.5** in

Terms of use · draft

This tool supports preliminary planning and does not replace review by a qualified professional. Verify inputs, applicable standards, connections, bearing, stability, and all scaffold-specific requirements before use. No certification or construction approval is provided.